MIL-E-17568C(NU)
24 January 1980
SUPERSEDING
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4 March 1974

MILITARY SPECIFICATION

# EMBROIDERY MATERIALS, METALLIC AND SYNTHETIC METALLIC

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope This specification covers requirements for metallic and synthetic metallic embroidery materials.

1.2 Classification Embroidery materials covered by this specification shall be of the following types and classes as specified (see 6.2):

| | |
|---|---|
| Type I | - Gold, defined as 24 karat gold over silver defined as 960 fine (minimum) |
| Type II | - Silver, defined as 960 fine (minimum) |
| Type III | - Synthetic Metallic |

* 1.2.1 Classes

| | |
|---|---|
| Types I and II | Class 1 - Matte bullion |
| | Class 2 - Shine bullion |
| | Class 3 - Light jaceron |
| | Class 4 - Heavy jaceron |
| | Class 5 - Frieze |
| | Class 6 - Lame |
| | Class 7 - Spangle |
| Type III | Class 8 - Gold synthetic |
| | Class 9 - Silver synthetic |
| | Class 10- Silver synthetic |
| | Class 11- Gold synthetic |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760, or by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

*2.1 Issues of documents The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

LAWS AND REGULATIONS

US POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications The following documents form a part of this specification to the extent specified herein. Unless a specific issue is indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Association, ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, ILL 60606.)

3. REQUIREMENT



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3.1 Guide sample Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from the specification may appear in the samples, in which case this specification shall govern.

*3.2 First article When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.2).

3.3 Materials

3.3.1 Type I - Gold The gold embroidery materials shall consist of 24 karat gold, plated to a silver wire core and shall conform to the requirements of table I. When tested as specified in 4.4.1, the gold shall be not less than 2.0% by weight of the total embroidery material and the silver shall be 960 fine (minimum).

3.3.2 Type II - Silver The silver embroidery materials shall be 960 fine (minimum), conforming to the requirements of Table I, when tested as specified in 4.4.2.

Table I - Construction and physical requirements

| | | Dimensions (inches) 1/ | | | | |
|---|---|---|---|---|---|---|
| Class | Material | Wire Type | Wire size 2/ | Wire tolerance | Finished material size | Material tolerance |
| 1 | Matte bullion | Round | .0038 | ±.0001 | .040 dia. | ± .005 |
| 2 | Shine bullion | Flat | .0027x.0055 | ±.0001 | .040 dia. | ± .005 |
| 3 | Light jaceron | Half Round | .0115 | ±.0001 | .036 dia. | ± .005 |
| 4 | Heavy jaceron | Half Round | .0135 | ±.0001 | .040 dia. | ± .005 |
| 5 | Frieze | Flat | .0027x.0055 | ±.0001 | .040 dia. | ± .005 |
| 6. | Lame | Flat | - | - | .091 wide | ± .001 |
| | | | | | .0017 thick | ± .0001 |
| 7. | Spangle | Round | - | - | .145 dia. | ± .015 |
| | | | | | .007 thick | ± .0001 |

1/ The size of round wires shall be the diameter, and of flat wires the thickness and width. The material size shall be the outside diameter except for lame and spangles. The size for the lame shall be the thickness and width. Classes 1, 2, 3, 4, and 5 shall be tightly coiled.

2/ All classes except class 7 are constructed from a single strand of wire.



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*3.3.3 Metallic yarn, synthetic (Type III) The synthetic metallic yarn shall consist of a synthetic metallic ribbon wound evenly and uniformly around a core yarn. The finished yarn for classes 8, 9, and 10 shall be size 225 (-5, +15) denier and the synthetic ribbon shall be wound with 26 (± 2) turns per inch. The finished yarn for Class 11 shall be size 400 (± 5) denier and the synthetic ribbon shall be wound with 44 (± 2) turns per inch. Testing shall be as specified in 4.3.2.

*3.3.3.1 Core yarn The core yarn for classes 8, 9 and 10 shall be of any suitable fiber, size or denier and tensile strength to produce a finish yarn suitable for Schiffli embroidery execution. The core yarn shall be of a color to match the colored synthetic ribbon cover. The core yarn for Class 11 shall be a 3 ply, 70 denier, continuous filament, white nylon.

*3.3.3.2 Synthetic metallic ribbon (gold, Class 8 and 11) The construction for the gold synthetic metallic ribbon shall be as specified in 3.3.3.2.1 and 3.3.3.2.2.

*3.3.3.2.1 Synthetic metallic ribbon (gold Class 8) The basic material shall be a metallic-silver polyester film, gold colored and cut to the specified width. The synthetic ribbon shall finish 1/69 to 1/64 inch wide and shall be 0.00075 (± 0.00005) inch thick. The ribbon shall consist of a gold colorant encased between a clear polyester film, 0.005 inch thick and a metallic-silver surface produced by the vacuum evaporation or deposition technique. The silver surface shall be backed with a substrate material suitable for the purpose intended. Testing shall be as specified in 4.3.2.

*3.3.3.2.2 Synthetic metallic ribbon (gold, Class 11) The synthetic metallic gold yarn shall be made of two, clear polyester films, 0.0005 (± 0.00005) inch thick. One surface shall be coated with a gold colorant. The gold colored surface shall then be subjected to a vacuum evaporation process in which a fine deposit of metallic aluminum is adhered to it. The metallized aluminum surface film is then laminated to the other clear polyester film with a polyester cross linking adhesive which contains a gold colorant, to form a single composite sheet 0.00100 (± 0.00005) inch thick. The metallic sheet shall be cut or slit into ribbons finishing 1/69 to 1/64 inch wide. Testing shall be as specified in 4.3.2.

*3.3.3.3 Synthetic metallic ribbon (silver, classes 9 and 10) The construction for the silver synthetic metallic ribbon shall be as specified in 3.3.3.3.1 or 3.3.3.3.2



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*3.3.3.3.1 Synthetic metallic ribbon (silver, class 9) The basic material shall be a metallic-aluminum polyester film laminate, cut to the specified width. The synthetic silver ribbon shall finish 1/69 to 1/64 inch wide and shall be 0.00075 (± 0.00005) inch thick. The ribbon shall consist of a metallic-aluminum encased or laminated between two clear polyester films with adhesives suitable to form a single composite sheet. The aluminum surface shall be produced by the vacuum evaporation or deposition technique. Testing shall be as specified in 4.3.2.

*3.3.3.3.2 Synthetic metallic ribbon (silver, class 10) The basic material shall be a metallic-silver polyester film cut to the specified width. The synthetic metallic ribbon shall finish 1/69 to 1/64 inch wide and shall be 0.00075 (± 0.00005) inch thick. The ribbon shall consist of pure silver encased between a clear polyester film, 0.0005 inch thick. The metallic silver surface shall be produced by the vacuum evaporation or deposition technique. The silver surface shall be backed with a substrate material suitable for the purpose intended. The core yarn used (see 3.3.3.1) shall be white. Testing shall be as specified in 4.3.2.

3.3.3.4 Colorfastness The combined type III core yarn and synthetic metallic ribbon cover shall show good colorfastness to dry cleaning, dry and wet heat, crocking, perspiration, bleaching, light, and weathering when tested as specified in 4.3.2.

3.3.3.5 Solvent resistance The synthetic metallic ribbon cover shall not show any evidence of discoloration or become tacky when tested as specified in 4.3.2.

3.3.4 Lacquer Unless otherwise specified (see 6.2), all classes for Types I and II shall be evenly and thoroughly lacquered with a pale, clear, synthetic lacquer. The use of pigmented lacquers are prohibited. The dry lacquer film shall be continuous, level, adherent, and essentially free from lint, dust and other foreign inclusions. The contractor shall furnish a Certificate of Compliance stating that the embroidery material has been lacquered (see 4.3.2.1).

3.5 Workmanship The finished item shall conform to the quality of product established by this specification. Occurrence of defects shall not exceed the applicable acceptable quality levels.

## 4. QUALITY ASSURANCE PROVISIONS



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4.1 Responsibility for inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 First article inspection The first article submitted in accordance with 3.2, shall be inspected as specified in 4.3.1.1 and 4.3.1.2 for compliance with design, construction, workmanship and dimensions.

4.3 Quality conformance inspection Inspection for quality conformance shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.3.1 Examination of the end item The defects found during the examination of the end item shall be classified in accordance with 4.3.1.1 and 4.3.1.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.3.3. The sample unit shall be one finished, 12 inch length for classes 1, 2, 3, 4, 5 and 6, 12 finished spangles for class 7 and one cone or bobbin of finished synthetic metallic material for classes 8, 9, 10 and 11. The lot size shall be expressed in terms of the sample unit.

4.3.1.1 Visual examination The finished embroidery material shall be examined and classified in accordance with the defects listed below.

| Examine | Defect |
|---|---|
| Type or class | Other than specified. |
| Color | 1. Not comparable to sample<br>2. Not color specified |
| Finish | 1. Gold plating chipped, tarnished, peeled, not continuous<br>2. Lacquer missing, tacky, not adherent<br>3. Dirt, grease, discolored |
| Material and construction | 1. Not coiled tightly (when specified)<br>2. Material or construction not as specified<br>3. Broken, cut, bent or damaged |



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4.3.1.2 Dimensional examination The finished Type I and Type II embroidery material shall be examined for defects in dimensions. Any measurement deviating from specified dimensions and tolerances specified in Table I, shall be scored a defect.

4.3.2 Testing of the end item The finished metallic and synthetic metallic embroidery material shall be tested for characteristics listed in Table II. The finished sample unit for testing, when applicable, shall be as follows: one 12 inch length for classes 1, 2, 3, 4, 5 and 6; 12 spangles for class 7; for type III, classes 8, 9. 10 and 11 one cone, spool or bobbin or sufficient amount to total a minimum of 360 yards. The lot size shall be expressed in terms of the sample unit. The lot shall be unacceptable if any sample unit fails to meet the requirements specified. The sample size shall be in accordance with the following:

| Lot size | Sample size (No. of sample units) |
|---|---|
| 800 or less | 2 |
| 801 thru 22,000 | 3 |
| 22,001 and over | 5 |

*Table II - Testing of end item

| Characteristic | Requirement paragraph | Test method |
|---|---|---|
| Synthetic yarn (combined metallic ribbon & core yarn) | | FED-STD-191 |
| Colorfastness to: | 1/ | |
| Dry cleaning | 3.3.3.4 | 5621 |
| Dry & wet heat (wool temp.) | 3.3.3.4 | 5640 |
| Crocking | 3.3.3.4 | 5651 |
| Perspiration | 3.3.3.4 | 5680 |
| Chlorine Bleach | 3.3.3.4 | 5600 |
| Light (40 hrs) | 3.3.3.4 | 5660 |
| Weathering (40 hrs) | 3.3.3.4 | 5671 |
| Durability of color | 3.3.3.5 | 4.4.4 |
| Tackiness | 3.3.3.5 | 4.4.5 |

1/ When specified, a certificate of compliance may be submitted for the colorfastness requirements and it shall be considered acceptable.



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*4.3.2.1 Certificate of compliance In addition to Table II the contractor shall submit a certificate of compliance for the materials and characteristics listed below:

| Component | Characteristics | Requirement paragraph |
|---|---|---|
| Cold for plating | Karat and weight | 3.3.1 1/ |
| Silver, wire core | Fineness | 3.3.1 2/ |
| Silver materials | Fineness, construction, and physical requirements | 3.3.2 3/ |
| Metallic yarn, snynthetic | Material identification | 3.3.3 |
| | Denier and turns per inch | 3.3.3 |
| Core yarn | Material identification and yarn size | 3.3.3.1 |
| Metallic ribbon, synthetic | Material identification, width, thickness, and construction | 3.3.3.2.1 or 3.3.3.2.2 or 3.3.3.3.1 or 3.3.3.3.2 |
| Lacquer | Type and proof of use | 3.3.4 |

1/ In the event of dispute the material shall be tested for gold content in accordance with 4.4.1.
2/ In the event of dispute the material shall be tested for silver fineness in accordance with 4.4.2
3/ In the event of dispute the material shall be tested for silver fineness in accordance with 4.4.3.

4.3.3 Inspection levels and acceptable quality levels The inspection levels and acceptable quality levels (AQL's) expressed as defects per one hundred units (DHU) for visual and dimensional examinations shall be as follows:

| | Inspection Level | AQL's |
|---|---|---|
| For visual and examination in 4.3.1.1 | I | 2.5 (one class) |
| For dimensional examination in 4.3.1.2 | S-3 | 2.5 (one class) |

4.4 Tests



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4.4.1 Type I - Gold content Boil 1 gram of gold embroidery material (accurately weighed) with 60 ml of 1:1 nitric acid ($HNO_3$) in a 400 ml beaker until all metals except gold are dissolved, and brown fumes are removed. Dilute to about 200 ml with hot water. Filter through a rapid, ashless filter paper into 1000 ml volumetric flask. Wash gold and paper with hot water. Reserve filtrate and washings for the analysis of silver (4.4.2). Wrap the washed filter paper with the gold in it (cutting off any excess filter paper) in a 2 inch by 2 inch piece of silver-free foil, and roll into a ball. Place the ball in a hollow of a bone ash cupel, and add silver-free lead shot around the ball. Place cupel in a muffle furnace for one hour at 1000 C. Remove from furnace and examine hollow for flakes of gold, impurities, and discolored gold bead. If any are noticed, add additional lead shot and return cupel to the muffle furnace for one hour. Cool the pure gold and weigh. The gold content shall be calculated as follows:

$$\% \text{ of 24 Karat gold} = \frac{\text{weight of gold x 100}}{\text{weight of sample}}$$

NOTE: Protect the solution from HCL fumes and use chloride - free water to prevent premature precipitation of silver chloride.

4.4.2 Type I - Silver fineness Take the filtrate and washings from 4.4.1, which are already in a 1000 ml volumetric flask, cool, and dilute to the mark. Pipette a 200 ml aliquot from the flask, into a 400 ml beaker. Heat to boiling and add a calculated amount of HCL (3:100) drop by drop from a burette while stirring (1 ml of 3:100 HCL solution precipitates about 0.038 grams of silver). Protect the silver chloride from light during the whole analysis. Avoid a large excess of HCL. When solution has cooled and precipitate has coagulated, determine whether precipitation is complete by adding a few more drops of HCL (3:100). If solution remains clear, filter by decantation into a weighed fritted glass crucible of medium porosity, using gentle suction. Wash the precipitate several times in the beaker with $HNO_3$ (1:200) and decant. Transfer the entire silver chloride precipitate to the crucible, and wash one time finally with cold water. Dry in an oven for not less than two hours, and weigh. The silver fineness shall be calculated as follows:

$$\text{Silver fineness} = \frac{\text{weight of silver chloride x 0.75263 x 5 x 1000}}{\text{weight of sample minus weight of gold}}$$



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4.4.3 Type II - Silver fineness Boil 1 gram of alloy (accurately we with 60 ml of 1:1 nitric acid ($HNO_3$) in a 400 ml beaker until all meta in solution and brown fumes are removed. Transfer quantitatively to a ml volumetric flask, cool, and dilute to the mark. Pipette a 200 ml a from the flask into a 400 ml beaker. Heat to boiling and add a calcul amount of HCL (3:100) drop by drop from a burette while stirring (1 ml 3:100 HCL solution precipitates about 0.038 grams of silver). Protect silver chloride from light during the whole analysis. Avoid a large e of HCL. When solution has cooled and precipitate has coagulated, dete whether precipitation is complete by adding a few more drops of HCL (3 If solution remains clear, filter by decantation into a weighed fritte crucible of medium porosity, using gentle suction. Wash the precipita several times in the beaker with $HNO_3$(1:200) and decant. Transfer the silver chloride precipitate to the crucible and wash one time finally cold water. Dry in an oven for not less than two hours, cool, and wei The silver fineness shall be calculated as follows:

$$\text{Silver fineness} = \frac{\text{weight of silver chloride x 0.75263 x 5 x 1000}}{\text{weight of sample}}$$

4.4.4 Durability of color A cloth saturated with methyl ethyl ketone solvent shall be lightly rubbed once, back and forth, over the synthet metallic yarn. Any evidence of discoloration shall be cause for failu

*4.4.5 Test for tackiness A cloth saturated with perchloroethylene so shall be lightly rubbed once, back and forth, over the synthetic metal yarn. Any evidence of tackiness shall be cause for failure.

5. PACKAGING

5.1 Preservation-packaging The finished metallic and synthetic meta embroidery materials shall be packaged to afford adequate protection a deterioration and damage during shipment from the supply source to the receiving activity. The contractor may use his standard commercial pra when it meets this requirement.

*5.2 Packing Finished metallic; synthetic metallic embroidery materi packaged as specified in 5.1, shall be packed in a manner to insure ca acceptance and safe delivery at destination at the lowest transportatic rate for such supplies. Containers shall comply with the US Postal Ser Manual, Uniform Freight Classification Rules, or National Motor Freight Classification Rules, as applicable.



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5.3 Marking Shipping containers and interior packages shall be marked in accordance with the contractor's standard practice or as specified by the procuring activity. As a minimum, container markings shall include the stock number, contractor's name, and contract number.

6. NOTES

6.1 Intended use The metallic and synthetic metallic embroidery materials covered by this specification are intended for use in the manufacture of embroidered insignia used by the Navy and Coast Guard.

6.2 Ordering data Procurement documents should specify the following:

a. Title, number and date of this specification
b. Types and classes required (see 1.2)
c. Availability of guide sample (see 3.1)
d. When a first article is required (see 3.2)
e. When lacquer is not required (see 3.3.4)

6.3 Samples For access to samples, address the procuring activity issuing the invitation for bids.

6.4 Changes from previous issue The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deltions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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